Phase 3

"CardKnowlogy"

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1. Design the Knowledge Base Structure

1.1 Facts (Working Memory)

Clinical Symptoms - Boolean

Shortness of breath → CF = 0.80
Orthopnea → CF = 0.85
Edema → CF = 0.75
Chest pain → CF = 0.90
Cough → CF = 0.60
Low activity → CF = 0.55
Palpitations → CF = 0.65
Dizziness → CF = 0.60
Syncope → CF = 0.95
Chest tightness → CF = 0.75

Vital Signs - boolean

Blood Pressure

≥ 180 → CF = 0.95
140–179 → CF = 0.70
< 90 → CF = 0.95

Heart Rate

120 → CF = 0.90
100–120 → CF = 0.75
< 50 → CF = 0.95

Respiratory Rate

22 → CF = 0.80

Oxygen Saturation

< 85 → CF = 0.98
85–90 → CF = 0.90
90–94 → CF = 0.75

Hemoglobin

< 10 → CF = 0.70

Temperature

38 → CF = 0.60

Background Information - Boolean

Hypertension → CF = 0.75
Diabetes → CF = 0.70
Heart disease → CF = 0.85
Obesity → CF = 0.65
Previous heart attack → CF = 0.95
Chronic lung disease → CF = 0.70
Kidney disease → CF = 0.75

Age - boolean

60 → CF = 0.70
40–60 → CF = 0.50
< 40 → CF = 0.30

1.2 Rules

The system knowledge is represented using production rules in the form of IF–THEN statements. Each rule defines a relationship between patient facts and a possible medical condition.

The IF part contains a combination of symptoms, vital signs, and background information.
The THEN part represents the inferred medical condition.

These rules collectively simulate the reasoning process of a medical expert by mapping input data to possible diagnoses.

1.3 Conditions - Output

Based on the applied rules, the system identifies possible medical conditions.

The system is capable of identifying the following conditions:

Acute Decompensated Heart Failure
Chronic Heart Failure (Stable)
Heart Failure with Preserved EF (HFpEF)
Acute Myocardial Infarction (STEMI / NSTEMI)
Acute Coronary Syndrome (ACS)
Stable Angina
Hypertensive Emergency
Atrial Fibrillation
Sudden Cardiac Arrest
Cardiogenic Shock

2-RULES

R0. IF recommendation exists THEN print disease, urgency, recommend CF = 1.0

Acute Decompensated Heart Failure (ADHF)

R1. IF shortness of breath AND orthopnea AND edema THEN disease = Acute Decompensated Heart Failure CF = 0.90

R2. IF shortness of breath AND edema AND oxygen saturation < 94 THEN disease = Acute Decompensated Heart Failure CF = 0.88

R3. IF orthopnea AND edema AND heart disease THEN disease = Acute Decompensated Heart Failure CF = 0.87

R4. IF shortness of breath AND respiratory rate > 22 AND edema THEN disease = Acute Decompensated Heart Failure CF = 0.85

R5. IF shortness of breath AND cough AND edema THEN disease = Acute Decompensated Heart Failure CF = 0.80

R6. IF Chronic Heart Failure AND oxygen saturation < 92 THEN disease = Acute Decompensated Heart Failure CF = 0.88

R7. IF Hypertensive Emergency AND shortness of breath THEN disease = Acute Decompensated Heart Failure CF = 0.82

R8. IF Dilated Cardiomyopathy AND edema AND shortness of breath THEN disease = Acute Decompensated Heart Failure CF = 0.90

R9. IF Chronic Heart Failure AND shortness of breath AND edema THEN disease = Acute Decompensated Heart Failure CF = 0.92

R10. IF disease = Acute Decompensated Heart Failure THEN urgency = HIGH CF = 1.0

R11. IF disease = Acute Decompensated Heart Failure THEN recommendation = "Seek urgent medical care and reduce fluid intake" CF = 1.0

Chronic Heart Failure (Stable)

R12. IF shortness of breath AND low activity AND edema THEN disease = Chronic Heart Failure CF = 0.80

R13. IF shortness of breath AND hypertension AND low activity THEN disease = Chronic Heart Failure CF = 0.78

R14. IF edema AND age > 60 AND hypertension THEN disease = Chronic Heart Failure CF = 0.82

R15. IF edema AND kidney disease AND hypertension THEN disease = Chronic Heart Failure CF = 0.80

R16. IF shortness of breath AND hemoglobin < 10 THEN disease = Chronic Heart Failure CF = 0.70

R17. IF disease = Chronic Heart Failure THEN urgency = MODERATE CF = 1.0

R18. IF disease = Chronic Heart Failure THEN recommendation = "Follow up with doctor and maintain medication compliance" CF = 1.0

Heart Failure with Preserved EF (HFpEF)

R19. IF shortness of breath AND hypertension AND age > 60 THEN disease = HFpEF CF = 0.82

R20. IF shortness of breath AND obesity AND hypertension THEN disease = HFpEF CF = 0.80

R21. IF shortness of breath AND diabetes AND age > 55 THEN disease = HFpEF CF = 0.78

R22. IF edema AND hypertension AND age > 60 THEN disease = HFpEF CF = 0.83

R23. IF disease = HFpEF THEN urgency = MODERATE CF = 1.0

R24. IF disease = HFpEF THEN recommendation = "Control blood pressure and schedule cardiology follow-up" CF = 1.0

Acute Myocardial Infarction (MI)

R25. IF chest pain AND chest tightness AND shortness of breath THEN disease = Acute Myocardial Infarction CF = 0.95

R26. IF chest pain AND heart rate > 100 AND oxygen saturation < 94 THEN disease = Acute Myocardial Infarction CF = 0.93

R27. IF chest pain AND diabetes AND hypertension THEN disease = Acute Myocardial Infarction CF = 0.88

R28. IF chest pain AND previous heart attack THEN disease = Acute Myocardial Infarction CF = 0.92

R29. IF chest pain AND syncope THEN disease = Acute Myocardial Infarction CF = 0.94

R30. IF Acute Coronary Syndrome AND heart rate > 110 THEN disease = Acute Myocardial Infarction CF = 0.85

R31. IF Acute Coronary Syndrome AND chest pain AND syncope THEN disease = Acute Myocardial Infarction CF = 0.90

R32. IF Hypertensive Emergency AND chest pain AND shortness of breath THEN disease = Acute Myocardial Infarction CF = 0.88

R33. IF disease = Acute Myocardial Infarction THEN urgency = CRITICAL CF = 1.0

R34. IF disease = Acute Myocardial Infarction THEN recommendation = "Go to emergency room immediately and avoid physical activity" CF = 1.0

Acute Coronary Syndrome (ACS)

R35. IF chest pain AND shortness of breath THEN disease = Acute Coronary Syndrome CF = 0.85

R36. IF chest tightness AND heart rate > 100 THEN disease = Acute Coronary Syndrome CF = 0.80

R37. IF chest pain AND diabetes THEN disease = Acute Coronary Syndrome CF = 0.78

R38. IF chest pain AND hypertension THEN disease = Acute Coronary Syndrome CF = 0.78

R39. IF disease = Acute Coronary Syndrome THEN urgency = HIGH CF = 1.0

R40. IF disease = Acute Coronary Syndrome THEN recommendation = "Seek urgent evaluation to rule out heart attack" CF = 1.0

Stable Angina

R41. IF chest pain AND low activity AND heart disease THEN disease = Stable Angina CF = 0.80

R42. IF chest tightness AND age > 50 THEN disease = Stable Angina CF = 0.75

R43. IF disease = Stable Angina THEN urgency = LOW CF = 1.0

R44. IF disease = Stable Angina THEN recommendation = "Rest and avoid exertion; schedule medical check-up" CF = 1.0

Hypertensive Emergency

R45. IF blood pressure ≥ 180 AND chest pain THEN disease = Hypertensive Emergency CF = 0.92

R46. IF blood pressure ≥ 180 AND shortness of breath THEN disease = Hypertensive Emergency CF = 0.90

R47. IF blood pressure ≥ 180 AND syncope THEN disease = Hypertensive Emergency CF = 0.95

R48. IF hypertension AND blood pressure ≥ 180 THEN disease = Hypertensive Emergency CF = 0.93

R49. IF disease = Hypertensive Emergency THEN urgency = HIGH CF = 1.0

R50. IF disease = Hypertensive Emergency THEN recommendation = "Seek immediate medical care to control blood pressure" CF = 1.0

Atrial Fibrillation

R51. IF palpitations AND heart rate > 100 THEN disease = Atrial Fibrillation CF = 0.85

R52. IF palpitations AND dizziness AND heart rate > 110 THEN disease = Atrial Fibrillation CF = 0.88

R53. IF palpitations AND syncope THEN disease = Atrial Fibrillation CF = 0.90

R54. IF disease = Atrial Fibrillation THEN urgency = MODERATE CF = 1.0

R55. IF disease = Atrial Fibrillation THEN recommendation = "Consult cardiologist for heart rhythm management" CF = 1.0

Sudden Cardiac Arrest

R56. IF syncope AND oxygen saturation < 85 THEN disease = Sudden Cardiac Arrest CF = 0.98

R57. IF syncope AND heart rate < 50 THEN disease = Sudden Cardiac Arrest CF = 0.97

R58. IF Acute Myocardial Infarction AND syncope AND heart rate < 50 THEN disease = Sudden Cardiac Arrest CF = 0.99

R59. IF Acute Decompensated Heart Failure AND oxygen saturation < 85 THEN disease = Sudden Cardiac Arrest CF = 0.97

R60. IF Cardiogenic Shock AND oxygen saturation < 85 THEN disease = Sudden Cardiac Arrest CF = 0.97

R61. IF disease = Sudden Cardiac Arrest THEN urgency = CRITICAL CF = 1.0

R62. IF disease = Sudden Cardiac Arrest THEN recommendation = "Call emergency services immediately and start CPR if trained" CF = 1.0

Cardiogenic Shock

R63. IF blood pressure < 90 AND heart rate > 120 THEN disease = Cardiogenic Shock CF = 0.95

R64. IF oxygen saturation < 85 AND blood pressure < 90 THEN disease = Cardiogenic Shock CF = 0.97

R65. IF Acute Decompensated Heart Failure AND blood pressure < 90 THEN disease = Cardiogenic Shock CF = 0.94

R66. IF Acute Myocardial Infarction AND blood pressure < 90 THEN disease = Cardiogenic Shock CF = 0.96

R67. IF Acute Decompensated Heart Failure AND oxygen saturation < 90 THEN disease = Cardiogenic Shock CF = 0.93

R68. IF Atrial Fibrillation AND heart rate > 120 AND dizziness THEN disease = Cardiogenic Shock CF = 0.90

R69. IF disease = Cardiogenic Shock THEN urgency = CRITICAL CF = 1.0

R70. IF disease = Cardiogenic Shock THEN recommendation = "Immediate emergency care required; patient in shock" CF = 1.0

3- Handling Multiple Possible Conditions & Conflicting conclusions

1. Building the Activated Rules Set (ARS)

When patient data is entered into the system, all rules are evaluated using forward chaining.

Every rule whose conditions are satisfied is activated and added to a set called:

Activated Rules Set (ARS)

2. Conflict Resolution Using LEX Strategy

Instead of firing all matched rules at once, the system selects which rule to execute using the **LEX strategy**, based on the following priorities:

2.1 Recency (Most Recent Facts First)

Rules that depend on the most recently added facts in the working memory are given higher priority.

2.2 Specificity (Complexity)

If multiple rules have the same recency, the system selects the rule that:

- Has more conditions in the IF part
- Or contains more specific conditions

2.3 Rule Order

If rules are still equal, the system selects the rule that appears first.

3. Rule Execution Process

- The system fires **one rule at a time** (not all at once)
- The inferred disease is added to working memory
- The system re-evaluates rules and repeats the process

This continues until:

- No more rules can fire
- Or a recommendation is generated (R0 is triggered)

4. Assigning Confidence Values (Certainty Factor - CF)

Each rule contributes a certainty factor (CF) to the inferred condition.

If multiple rules infer the same condition, their CFs are combined:

$$CF_{combined} = CF1 + CF2 - (CF1 \times CF2)$$

Example

If two rules infer Acute Decompensated Heart Failure:

- $CF1 = 0.90$
- $CF2 = 0.88$
- $CF \approx 0.90 + 0.88 - (0.90 \times 0.88) = 0.988 \approx 0.99$

5. Final Output

The system produces:

- Primary Condition
- Confidence (CF)
- Urgency Level
- Recommendation

Example:-

1. Initial Facts (Working Memory)

The user provides symptoms with their initial Certainty Factors:

- **F1:** Heart disease (CF = 0.85) - *Oldest*
- **F2:** Edema (CF = 0.75)
- **F3:** Shortness of breath (CF = 0.80)
- **F4:** Orthopnea (CF = 0.85) - *Newest*

2. Execution Trace with LEX and CF Combining

Cycle 1

- **Conflict Set:** {**R1**, **R3**}
- **Resolution (LEX): R1** wins (Recency: F3 is newer than F1).
- **Calculation:** Min(Inputs) = Min(F2, F3, F4) = 0.75
    - CF_{R1} = 0.90 * 0.75 = 0.675
- **Action:** Add **F5: Disease = ADHF** (CF = 0.675).

Cycle 2

- **Conflict Set:** {**R10**, **R11**, **R3**}
- **Resolution (LEX): R10** wins (Recency: F5 is the newest; R10 comes first in **Order**).
- **Action:** Add **F6: Urgency = HIGH**.

Cycle 3

- **Conflict Set:** {**R11**, **R3**}
- **Resolution (LEX): R11** wins (Recency: F5 is newer than facts in R3).
- **Action:** Add **F7: Recommendation exists**.

Cycle 4

- **Conflict Set:** {**R0**, **R3**}
- **Resolution (LEX): R0** wins (Recency: It matches **F7**, the latest fact).
- **Action: Print Initial Output.** (At this point, Disease CF is 0.675).

Cycle 5

- **Conflict Set:** {**R3**}
- **Resolution:** R3 finally fires.
- **Calculation:** Min(Inputs) = Min(F1, F2, F4) = 0.75
  - CF_{R3} = 0.87 * 0.75 = 0.6525
- **Combining CFs:** Since both are positive:
  CF= (0.675+0.6525)-(0.675*0.6525)= 0.887

- **Action:** Update **F5: Disease = ADHF** with CF=0.887.

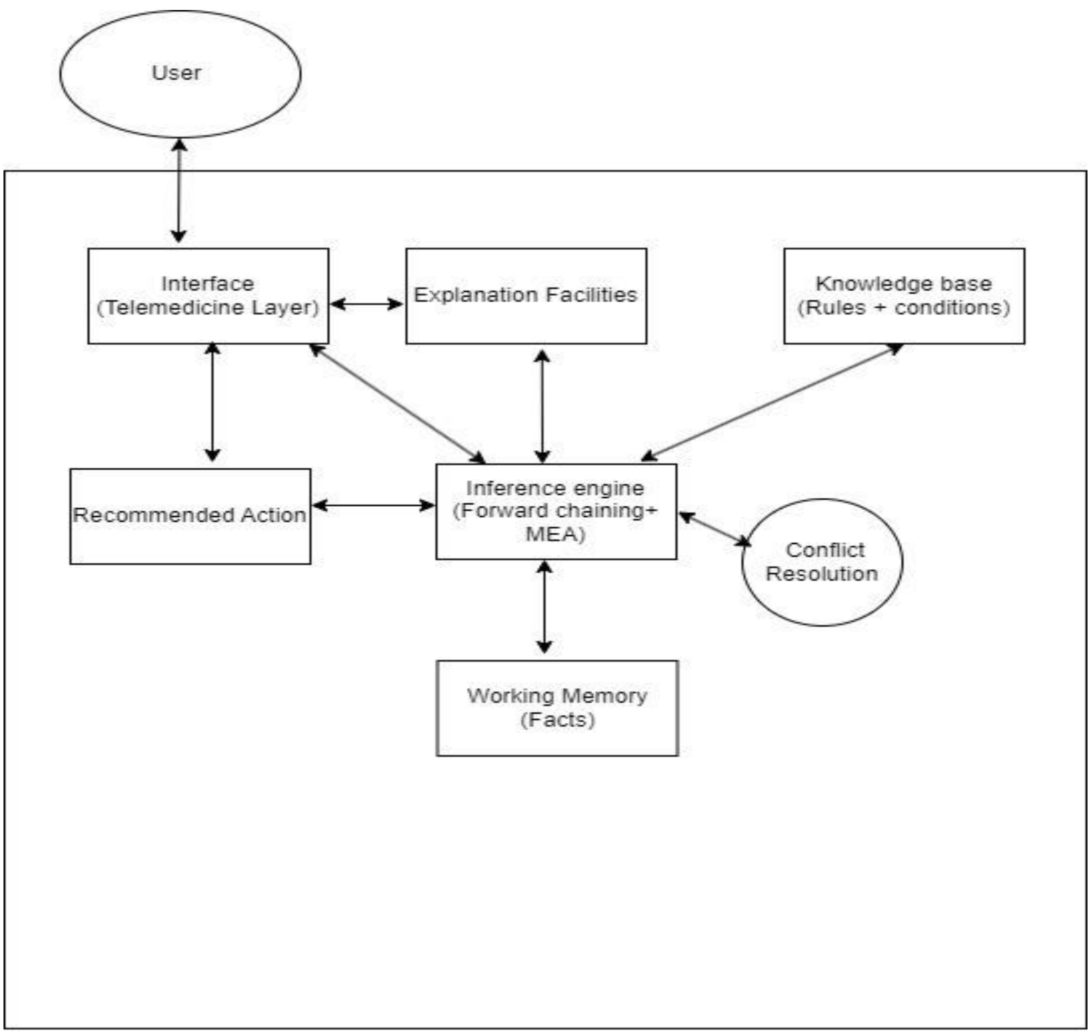
Final System Output

Parameter	**Result**
Inferred Disease	Acute Decompensated Heart Failure
Urgency Level	HIGH
Medical Advice	Seek urgent medical care and reduce fluid intake

4- KBS System architecture

➔ The Heart Failure Telemedicine Triage System is a rule-based expert system designed for early cardiac assessment in resource-limited settings (e.g., rural Egypt). It follows the classical expert system architecture with enhancements for medical safety, multiple diagnoses, and telemedicine usability.

The system consists of the following main components:



1.1 User Interface

- **Allows the user (patient, nurse, general practitioner, or hotline operator) to input patient data easily through a telemedicine-friendly interface.**
- **Inputs include:**
  - **Clinical Symptoms (Boolean values with associated Certainty Factors)**
  - **Vital Signs (boolean values with predefined thresholds and Certainty Factors)**
  - **Background Information (Boolean values with Certainty Factors)**
- **Displays the final output in clear, simple, and actionable language suitable for telemedicine settings.**
- **Includes a strong medical disclaimer: "This system is designed to support early assessment and does not replace professional medical diagnosis or consultation with a cardiologist."**

1.2 Working Memory (Facts / Dynamic Database)

- **Stores all current case-specific patient data entered through the User Interface.**
- **Acts as short-term memory during the reasoning process, holding asserted facts, intermediate inferences, and fired rule results.**
- **Each fact is associated with a Certainty Factor (CF) that reflects its clinical importance or severity.**

Categories of Facts:

a. Clinical Symptoms (Boolean + CF)

- **Shortness of breath (CF = 0.80)**
- **Orthopnea (CF = 0.85)**
- **Edema (CF = 0.75)**
- **Chest pain (CF = 0.90)**
- **Cough (CF = 0.60)**
- **Low activity (CF = 0.55)**
- **Palpitations (CF = 0.65)**
- **Dizziness (CF = 0.60)**
- **Syncope (CF = 0.95)**
- **Chest tightness (CF = 0.75)**

b. Vital Signs (Numeric with thresholds and CF)

- **Blood Pressure: ≥ 180 (CF = 0.95), 140–179 (CF = 0.70), < 90 (CF = 0.95)**
- **Heart Rate: > 120 (CF = 0.90), 100–120 (CF = 0.75), < 50 (CF = 0.95)**
- **Respiratory Rate: > 22 (CF = 0.80)**
- **Oxygen Saturation: < 85 (CF = 0.98), 85–90 (CF = 0.90), 90–94 (CF = 0.75)**
- **Hemoglobin: < 10 (CF = 0.70)**
- **Temperature: > 38 (CF = 0.60)**

c. Background Information (Boolean + CF)

- **Hypertension (CF = 0.75), Diabetes (CF = 0.70), Heart disease (CF = 0.85), Obesity (CF = 0.65), Smoking (CF = 0.60), Family history (CF = 0.55), Previous heart attack (CF = 0.95), Chronic lung disease (CF = 0.70), Kidney disease (CF = 0.75)**
- **Age: > 60 (CF = 0.70), 40–60 (CF = 0.50), < 40 (CF = 0.30)**

Facts in Working Memory are dynamically updated as new inferences are made during forward chaining.

1.3 Knowledge Base

- **Contains the static domain expertise encoded by medical experts.**
- **Includes:**
  - **Production Rules (IF–THEN statements): 70+ rules (R0–R70) with associated Certainty Factors (CF).**
  - **List of 10 possible cardiac conditions.**
  - **Urgency levels and corresponding recommended actions defined within the rules.**
  - **Clinical thresholds for vital signs.**
  - **Predefined Certainty Factors (CF) for facts and rules.**
- **The Knowledge Base is completely separated from the inference mechanism, allowing easy updating and expansion by cardiologists.**

1.4 Inference Engine

- **The core reasoning component ("brain") of the expert system.**
- **Uses Forward Chaining (data-driven reasoning): Starts with facts in the Working Memory and repeatedly applies matching production rules to derive new facts and possible conditions.**
- **Operates in a recognize-act cycle: Match rules → Fire rule → Update Working Memory.**
- **After rule firing, passes all inferred conditions and their associated Certainty Factors to the Conflict Resolution & Prioritization Module.**

1.5 Conflict Resolution

- **The system applies a strict LEX-based hierarchy to prioritize and aggregate medical evidence:**
- **Recency (Primary Filter): Rules matching the newest clinical data (e.g., current symptoms) fire before rules matching older data (e.g., medical history).**
- **Evidence Aggregation: When multiple rules identify the same condition, the CF is updated incrementally:**
- **CF_{combined} = CF_{old} + CF_{new} - (CF_{old} * CF_{new})**
- **Complexity (Tie-breaker): If recency is equal, the most specific rule (highest number of IF conditions) fires first.**
- **Primary Diagnosis: The condition with the highest final CF is selected as the primary result.**
- **Output Execution: Urgency and Recommendation rules fire last, as they depend on the newly inferred diagnosis, followed by the final print command (R0).**

1.6 Explanation Facility

- **Provides transparency by explaining why a particular conclusion was reached.**
- **Generates:**
  - **The Primary Condition with its final Certainty Factor (CF).**
  - **List of key fired rules and triggering facts.**
  - **Clinical Notes (e.g., "Based on: shortness of breath + edema + hypertension + age > 60").**
  - **Confidence interpretation of the CF value.**
- **Builds user trust and supports educational use by clinicians and students.**

1.7 Recommended Action Generator

- **Takes the output from the Conflict Resolution & Prioritization Module (Primary Condition, Urgency Level, and final CF).**
- **Maps the urgency level to a clear, specific Recommended Action as defined in the rules (e.g., R11, R18, R24, R34, R50, etc.).**
- **Examples of mapping:**
  - **CRITICAL → "Go to the emergency room immediately"**

- **HIGH → "Seek urgent medical care"**
- **MODERATE → "Follow up with doctor within the next few days"**
- **LOW → "Schedule a routine follow-up and rest"**

- **Always driven by the primary condition and its associated urgency (Safety-First principle).**

2. Data Flow in the Architecture

1. **User enters patient data via the User Interface → facts (with CF) are stored in Working Memory.**
2. **Inference Engine (Forward Chaining) evaluates rules from the Knowledge Base and fires applicable rules.**
3. **Fired rules populate the Possible Conditions Set (PCS) with their individual CF values.**
4. **Conflict Resolution (by LEX)Prioritization Module removes duplicates, combines CF values, ranks conditions, and selects the Primary Condition.**
5. **Recommended Action Generator determines the appropriate action based on the urgency of the primary condition.**
6. **Explanation Facility generates the final structured output including Primary Condition, CF, Urgency, Recommended Action, and Clinical Notes.**
7. **The complete result is displayed to the user through the User Interface.**

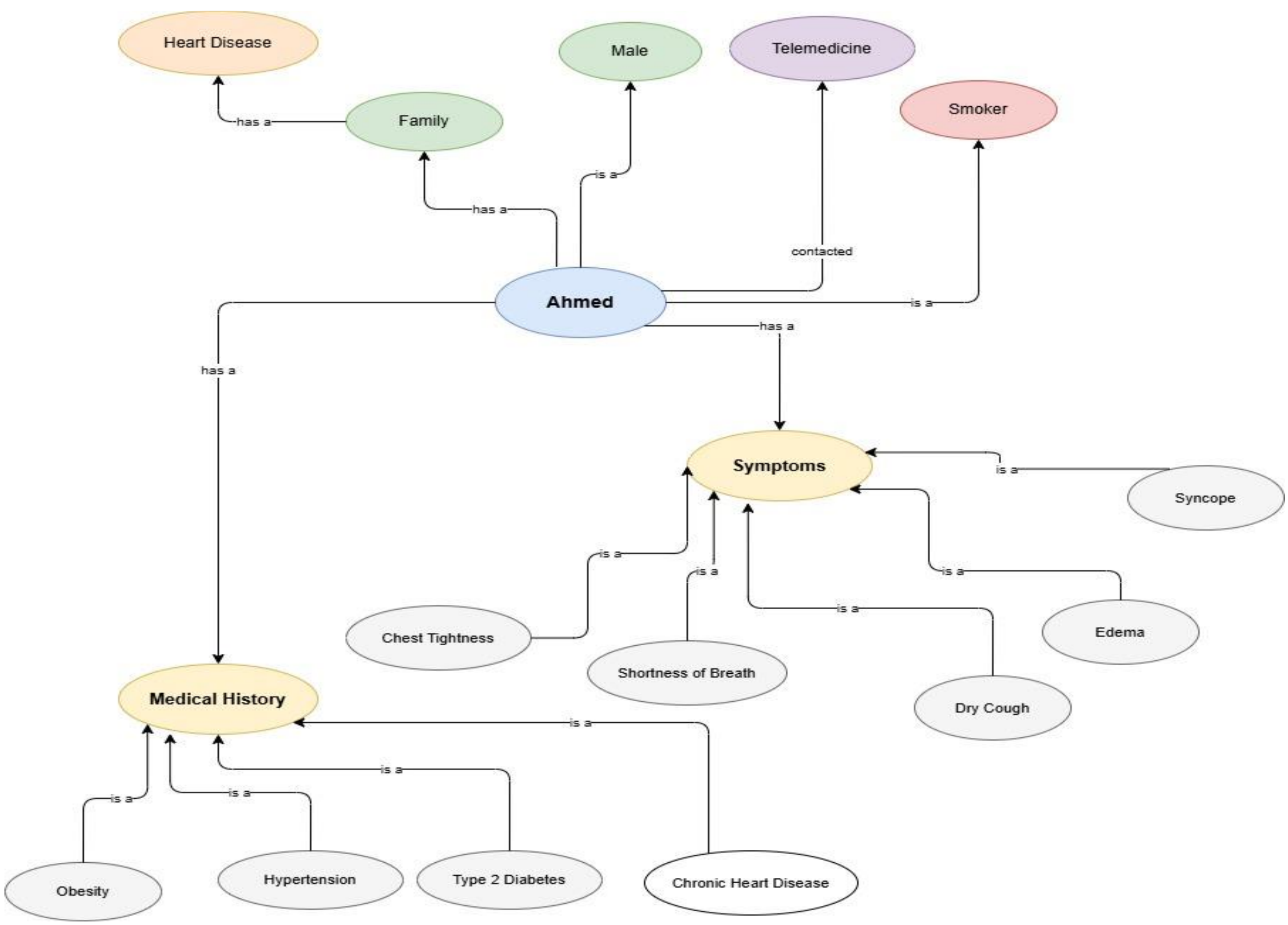
Scenario:

Mr. Ahmed Hassan is a 67-year-old male patient living in a rural area near Alexandria, Egypt. He contacted the telemedicine hotline complaining of several symptoms. He is experiencing shortness of breath that worsens when lying flat (orthopnea), noticeable swelling in both legs and ankles (edema), mild chest tightness, a dry cough especially at night, significantly reduced activity tolerance (he gets tired after walking only a short distance), and occasional dizziness. He denies palpitations and fainting (syncope).

During the call, his vital signs were recorded as follows: blood pressure of 162/98 mmHg, heart rate of 108 beats per minute, respiratory rate of 24 breaths per minute, oxygen saturation of 91%, hemoglobin level of 10.8 g/dL, and normal body temperature of 37.2°C.

From his medical background, Mr. Hassan has a known history of hypertension for the past four years with irregular medication intake, type 2 diabetes, chronic heart disease, and obesity with a BMI of 32. He is a current smoker consuming about 10 cigarettes per day for the last 30 years. He also has a positive family history of heart disease.

This combination of symptoms, abnormal vital signs, and multiple risk factors led the system to infer multiple possible cardiac conditions, with Acute Decompensated Heart Failure emerging as the strongest diagnosis.



5- Inference Process Flow & Explanation System

1. Inference Process Flow

- The system utilizes **Forward Chaining** (Data-Driven) to move from patient symptoms to a final triage decision.

Step 1: Fact Acquisition

- All inputs (Symptoms, Vitals, Background) are stored in **Working Memory**.
- Each fact is assigned an initial **CF** and a **Timestamp**.

Step 2: Rule Matching (Conflict Set)

- The Inference Engine scans the Knowledge Base.
- All rules whose IF conditions are met are added to the **Conflict Set**.

Step 3: Conflict Resolution (LEX Strategy)

- Before firing, the engine prioritizes the Conflict Set using the **LEX hierarchy**:
- **Recency:** Rules using the most recently entered facts fire first.
- **Complexity:** If recency is tied, rules with the most conditions (most specific) fire first.
- **Order:** If both are tied, the rule appearing first in the Knowledge Base fires.

Step 4: Rule Firing & Evidence Aggregation

- The prioritized rule fires, producing an inferred condition.
- **PCS Construction:**
    - If the condition is new: It is added to the **Possible Conditions Set (PCS)**.
    - If the condition already exists: The system aggregates the evidence using:

  $CF_{combined} = CF_{old} + CF_{new} - (CF_{old} * CF_{new})$
- **Loop:** The system returns to Step 2 to re-match based on the newly inferred data.

Step 5: Primary Condition Selection

- Once no more rules can fire, the engine stops.
- The system ranks all conditions in the **PCS** by their final **Combined CF**.
- **Primary Condition** = The condition with the **highest CF**.
- *Clinical Safety:* This avoids "rule-order bias" by ensuring the diagnosis with the strongest cumulative evidence wins.

Step 6: Output Generation

- The Primary Condition triggers final rules for **Urgency Level** and **Recommended Action**.
- **Final Output Includes:**
    - Primary Diagnosis & Final Confidence (CF).
    - Urgency Level (Critical, High, Moderate, Low).
    - Specific Medical Instructions.